

A 96-element Sidon subset of the seventeenth multinomial coefficient layer

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Abstract

We exhibit 96 distinct multinomial coefficients associated with partitions of 17 whose 4,656 unordered pair sums, including doubles, are all distinct. This proves $s(17) \geq 96$, increasing by seven the lower bound reported in Huber's arXiv:2608.02667v2 and OEIS A398429. We provide the complete integer set, one partition certificate for each element, and a short exact verifier. The problem of determining the exact value of $s(17)$ remains unresolved.

For an integer partition $\lambda = (\lambda_1 \geq \dots \geq \lambda_t \geq 1)$ of n , put

$$m_\lambda = \frac{n!}{\lambda_1! \dots \lambda_t!}, \quad \mathcal{M}_n = \{m_\lambda : \lambda \vdash n\}.$$

The set $\mathcal{M}_n$ contains each integer value once, even if several partitions give the same value. A subset S is *Sidon* if $a + b = c + d$ with $a, b, c, d \in S$ implies equality of the unordered pairs $\{a, b\} = \{c, d\}$. Equal summands are allowed. Define $s(n) = \max\{|S| : S \subseteq \mathcal{M}_n \text{ is Sidon}\}$.

Huber reports a set of size 89 in $\mathcal{M}_{17}$ without an optimality claim [1, Section 14.1]; Question 14.3 in Section 14.4 asks to determine $s(17)$. The inspected OEIS record also gives the lower bound 89 [2]. The following construction improves this reported bound.

Theorem 1. *The 96 integers in Table 1 form a Sidon subset of $\mathcal{M}_{17}$. In particular, $s(17) \geq 96$.*

Proof. Let S be the set of entries in the two value columns of Table 1. The entries are distinct. Each accompanying partition $\lambda(v)$, with $t(v)$ parts, has sum 17, and exact factorial arithmetic gives

$$v \prod_{j=1}^{t(v)} \lambda_j(v)! = 17! = 355687428096000 \quad (v \in S).$$

Thus $S \subseteq \mathcal{M}_{17}$. There are $96 \cdot 97/2 = 4656$ unordered pairs with repetition from S . Computing their integer sums gives 4,656 distinct values. The complete check is reproduced by the code on page 3, using the accompanying table data. Consequently no two different unordered pairs have the same sum, including pairs with equal summands. This proves the Sidon property and the bound. $\square$

This is a finite existence result. It neither establishes $s(17) = 96$ nor excludes a set of size 97. The comparison is limited to the identified source records inspected on 8 September 2026; no exhaustive priority claim is made. The contribution is the larger explicit construction, rather than a different criterion for testing the Sidon property.

The integer certificate

In Table 1, an exponent denotes the multiplicity of a part: for example, $(15, 1^2)$ denotes $(15, 1, 1)$. Read the left value column from top to bottom, followed by the right value column, to obtain the increasing list of all 96 values. The supplementary file `certificate.json` gives these same value-partition pairs.

Table 1: The 96 values and their partition certificates.

v	$\lambda(v)$	v	$\lambda(v)$
1	(17)	343062720	$(6^2, 2, 1^3)$
17	(16, 1)	490089600	$(9, 2, 1^6)$
136	(15, 2)	551350800	$(8, 2^4, 1)$
272	$(15, 1^2)$	686125440	$(6^2, 1^5)$
680	(14, 3)	1072071000	$(4^4, 1)$
2040	(14, 2, 1)	1960358400	$(7, 3^2, 1^4)$
4080	$(14, 1^3)$	2144142000	$(4^3, 3, 2)$
6188	(12, 5)	2287084800	$(6, 3^3, 1^2)$
9520	(13, 3, 1)	3087564480	$(5^2, 2^3, 1)$
14280	$(13, 2^2)$	4410806400	$(8, 2, 1^7)$
19448	(10, 7)	5145940800	$(6, 4, 2^2, 1^3)$
24310	(9, 8)	6175128960	$(5^2, 2^2, 1^3)$
28560	$(13, 2, 1^2)$	6432426000	$(4^3, 2^2, 1)$
30940	(12, 4, 1)	8821612800	$(8, 1^9)$
57120	$(13, 1^4)$	11435424000	$(4, 3^4, 1)$
74256	(11, 5, 1)	12350257920	$(5^2, 2, 1^5)$
123760	$(12, 3, 1^2)$	12864852000	$(4^3, 2, 1^3)$
136136	(10, 6, 1)	17643225600	$(7, 2^2, 1^6)$
218790	$(8^2, 1)$	22870848000	$(3^5, 2)$
247520	$(11, 3^2)$	24700515840	$(5^2, 1^7)$
742560	$(12, 1^5)$	35286451200	$(7, 2, 1^8)$
816816	$(10, 5, 1^2)$	38594556000	$(4^2, 2^4, 1)$
875160	(8, 7, 2)	45741696000	$(3^5, 1^2)$
1113840	$(11, 2^3)$	70572902400	$(7, 1^{10})$
1701700	$(9, 4^2)$	77189112000	$(4^2, 2^3, 1^3)$
1750320	$(8, 7, 1^2)$	82335052800	$(6, 3, 1^8)$
2227680	$(11, 2^2, 1^2)$	137225088000	$(3^4, 2, 1^3)$
2333760	$(7^2, 3)$	185253868800	$(5, 2^4, 1^4)$
3063060	(8, 5, 4)	231567336000	$(4, 2^6, 1)$
4455360	$(11, 2, 1^4)$	274450176000	$(3^4, 1^5)$
4900896	$(7, 5^2)$	370507737600	$(5, 2^3, 1^6)$
5717712	$(6^2, 5)$	494010316800	$(6, 1^{11})$
7001280	$(7^2, 2, 1)$	741015475200	$(5, 2^2, 1^8)$
8910720	$(11, 1^6)$	823350528000	$(3^3, 2, 1^6)$
10210200	$(9, 4, 2^2)$	1389404016000	$(2^8, 1)$
13613600	$(9, 3^2, 2)$	1646701056000	$(3^3, 1^8)$
14002560	$(7^2, 1^3)$	1852538688000	$(4, 2^3, 1^7)$
24504480	$(10, 2^2, 1^3)$	2778808032000	$(2^7, 1^3)$
27227200	$(9, 3^2, 1^2)$	4940103168000	$(3^2, 2, 1^9)$
34306272	$(6, 5^2, 1)$	5557616064000	$(2^6, 1^5)$
61261200	$(9, 2^4)$	9880206336000	$(3^2, 1^{11})$
81681600	$(9, 3, 2, 1^3)$	11115232128000	$(2^5, 1^7)$
91891800	$(8, 4, 2^2, 1)$	22230464256000	$(2^4, 1^9)$
102918816	$(5^3, 2)$	29640619008000	$(3, 2, 1^{12})$
114354240	$(6^2, 3, 1^2)$	44460928512000	$(2^3, 1^{11})$
205837632	$(5^3, 1^2)$	88921857024000	$(2^2, 1^{13})$
214414200	$(5, 4^3)$	177843714048000	$(2, 1^{15})$
326726400	$(7, 3^3, 1)$	355687428096000	(1^{17})

Exact verification and reproducibility

Place the complete program `verify.py` beside `certificate.json` and run `python3 verify.py`. It uses only standard-library, arbitrary-precision integer arithmetic. It checks every membership identity and unordered pair, including all 96 doubles, and also checks that the $96 \cdot 95/2 = 4560$ positive differences are distinct.

```
import json
from itertools import combinations, combinations_with_replacement
from math import factorial, prod
from pathlib import Path

def require(condition):
    if not condition:
        raise ValueError("Certificate check failed")

rows = json.loads(Path(__file__).with_name("certificate.json").read_text())
values = [row["value"] for row in rows]
require(all(type(v) is int and v > 0 for v in values))
require(values == sorted(set(values)) and len(values) == 96)
for row in rows:
    p = row["partition"]
    require(all(type(k) is int and k > 0 for k in p))
    require(p == sorted(p, reverse=True) and sum(p) == 17)
    require(row["value"] * prod(factorial(k) for k in p) == factorial(17))
sums = [a + b for a, b in combinations_with_replacement(values, 2)]
require(len(sums) == len(set(sums)) == 4656)
differences = [b - a for a, b in combinations(values, 2)]
require(len(differences) == len(set(differences)) == 4560)
print("96 memberships; 4656 distinct pair sums; 4560 distinct differences.")
```

The output is

```
96 memberships; 4656 distinct pair sums; 4560 distinct differences.
```

The membership identities avoid any dependence on complete enumeration of $\mathcal{M}_{17}$. Verification requires no search solver or optimality computation.

Computational context. An iterated greedy search produced the set; verification of its output is independent of those search decisions. Zeng’s strict-pair product problem [3] provides adjacent context for certifying a finite set by checking pairwise arithmetic values. It concerns products of distinct elements and supplies no theorem-level dependency for this additive result.

Assistance and scope. AI assisted the search implementation, verification, and manuscript preparation. A separate AI review pass recomputed the membership and pair-sum checks with a different implementation; this is not external peer review. The displayed certificate supports only Theorem 1.

References

- [1] Felix Huber, *Large Sidon Subsets and Pair-Sum Multiplicities of Distinct Multinomial Coefficients*, arXiv:2608.02667v2 (2026). <https://arxiv.org/abs/2608.02667v2>.
- [2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A398429, <https://oeis.org/A398429> (inspected 8 September 2026).
- [3] Zijian Zeng, *The Strict-pair Product Extremal Value at 62 is 31* (2026). <https://doi.org/10.2139/ssrn.7380458>.